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A SIMPLE 3-CENTIMETER–BAND PULSE OSCILLATOR FOR AN ECHO-SPECTROMETER

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A 3-centimeter band oscillator is described. The oscillator is of a waveguide design with power combining using AA715D Gunn diodes. The results of experimental studies of its operation in the pulsed mode are given. At a frequency of 9.4 GHz, a pulse power of 3 W was achieved. The oscillator has been tested in an electron spin-echo spectrometer.

In electron spin-echo spectrometers, either a magnetron or a low-power klystron, whose signal is amplified by a high-power traveling-wave tube, is commonly used as a microwave oscillator. However, these devices require a high-voltage (1–10 kV) power supply. In addition, a magnetron transmitter requires a complex system to form short (10–200 nsec) high-voltage modulation pulses. On the other hand, in many cases an oscillator power of 100 W is sufficient to excite a spin-echo signal in the sample placed inside a 3-centimeter–band rectangular resonator with the H_{102} wave mode [1]. The application of loop-gap resonators allows one to reduce the required power down to units of watts [2]. In this case one can implement a semiconductor microwave oscillator; for instance, using high-power Gunn diodes [3]. In comparison with traditional oscillators used in electron spin-echo spectrometers, the main advantages of a diode oscillator are its simplicity and low-voltage power supply. Besides, such oscillators can be synchronized with an external continuous-wave low-power microwave oscillator, which permits one to obtain coherent pulse sequences [4].

In this article a waveguide implementation of a 3-centimeter–band oscillator with power combining using commercially available AA715D diodes is described (see Fig. 1). The diodes 1 are set coaxially at the center of a 23 × 5 mm waveguide section 2 with length $\lambda/2 = 22$ mm. The diodes are sealed inside copper rods 3, which are screwed into the wide walls of the waveguide. Between the heads of the diodes a copper strip 4 is fixed, pulsed power being supplied to it through a plug 5 filled with ferroepoxy. A piston 6 in a stabilizing resonator 7 is used for mechanical frequency tuning. The cross-section of the resonator is 23 × 10 mm. A $\lambda/4$-long matching element 8 with a capacitive screw 9 and a diaphragm 10 serve the purpose of matching the oscillator to the load. All waveguide parts are made of brass. A main resonator 2 is made massive for better heat removal.

A schematic diagram of the pulse modulator is shown in Fig. 2. It permits the formation of a voltage pulse of 25 V in amplitude in a load of 0.5 Ω. The resistance of the high-power Gunn diode over the linear part of its volt-ampere characteristic is 0.6–2.5 Ω. Circuit R*C* is devised to suppress parasitic oscillations that arise in the diodes' supply circuit. We set R^* at about 3–10 Ω and C^* at 1–20 nF, although the time constant of the circuit should be as small as possible. The modulator is driven by a G5–66 oscillator.

At an operating frequency of 9.4 GHz, a pulse power of 3 W was reached with a modulation voltage of 20–25 V (in the continuous-wave mode, it was 0.9 W). Without a fall in power, one can vary the duration of the microwave pulses from 30 nsec to 1 msec and decrease the pulse period-to-pulse duration ratio down to 200. The rise and fall times of the pulses are 10 nsec. With the frequency tuned within the range 9.2–9.6 GHz, the power diminishes by 1 dB. By means of an S4-27 spectrum analyzer, an assessment was made of an unwanted deviation of the generated oscillation frequency caused by temperature variations during the modulation pulse. A substantial spectrum broadening is observed for pulses with $\tau > 10$ μsec, and the −6 dB half-bandwidth of the oscillations is 4 MHz at $\tau = 10$ μsec. At $\tau < 1$ μsec, the frequency deviation is negligible. When generating a pair of pulses delayed by 0.05–100 μsec, the difference in the carrier frequency between the pulses is less than ≤ 1 MHz.

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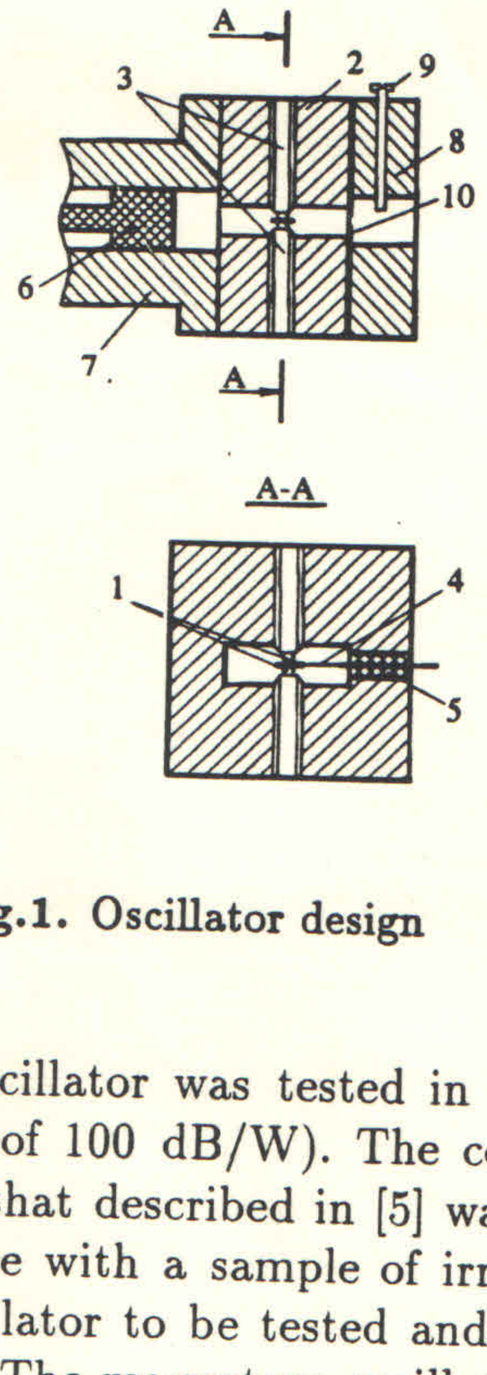


Fig.1. Oscillator design

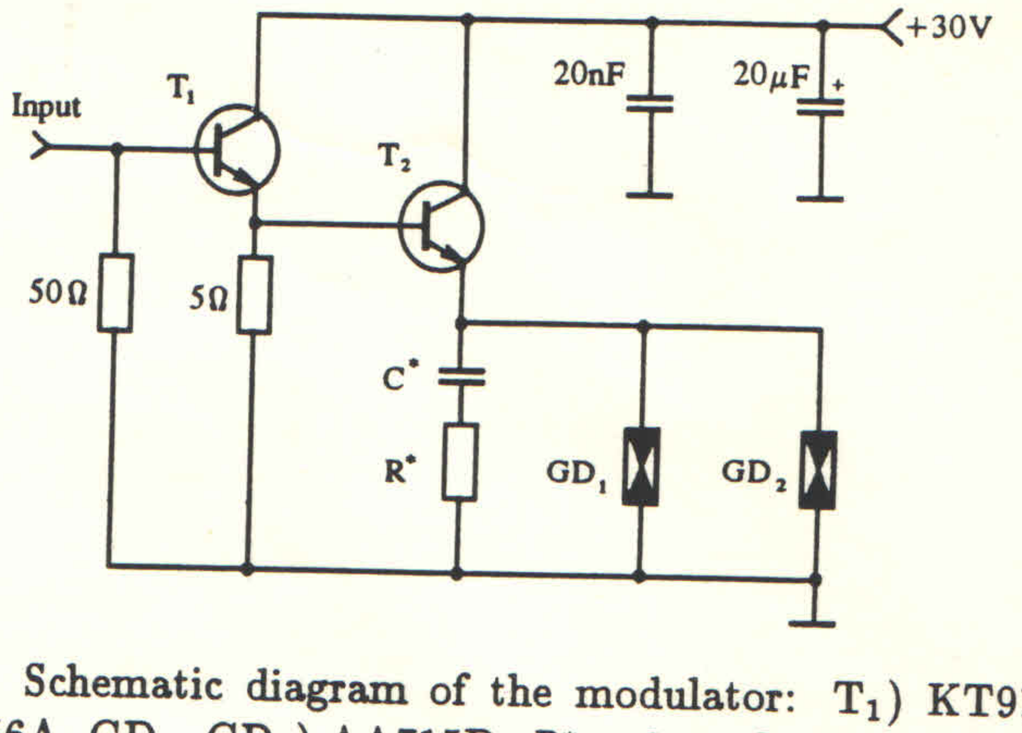


Fig.2. Schematic diagram of the modulator: T_1) KT913V, T_2) KT956A, GD_1, GD_2) AA715D, $R^* = 3$–$10\ \Omega$, $C^* = 1$–20 nF.

The oscillator was tested in an electron spin-echo apparatus with a direct-detection receiver (with a sensitivity of 100 dB/W). The center frequency of the spectrometer was 9.4 GHz. A loop-gap resonator similar to that described in [5] was employed as a measuring cell. The experiment was carried out at room temperature with a sample of irradiated quartz with a spin concentration of 10^{17} cm^{-3}. The capabilities of the oscillator to be tested and of a magnetron transmitter for electron spin-echo signal excitation were compared. The magnetron oscillator provided pulses with a fixed duration of 100 nsec and a power of 1 kW. To observe the peak double-pulse echo signal ($120° + 120°$) its power had to be diminished by 30 dB. Under the same conditions, the Gunn-diode oscillator enabled us to obtain a ($120° + 120°$)-echo with an excitation-pulse duration of 200 nsec. The power margin proved to be 3 dB.

The study revealed that pulse oscillators using Gunn diodes with power combining can be successfully utilized in electron spin-echo spectrometers. The use of such semiconductor oscillators along with measuring cells such as loop resonators and with modern microwave GaAs FET amplifiers greatly simplifies the design of an electron spin-echo spectrometer, lowers the power consumption, and reduces its dimensions, thereby making the spectrometers available for many researchers.

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